

Introduction to Data Science (CS2205)

Full Marks: 30

Time: 2 Hours

Attempt All Questions.

1. Choose the correct answer.

(10)

- i) What is the main drawback of mean imputation for missing numerical data?
- a. It increases data variance
 - b. It removes outliers
 - c. It distorts the data distribution
 - d. It increases dimensionality
- ii) What is the primary objective of data normalization in machine learning?
- a. Reduce dataset size
 - b. Bring features to a common scale
 - c. Improve model interpretability
 - d. Remove redundant features
- iii) Which problem is most likely to occur if data integration is poorly handled?
- a. Overfitting
 - b. Data redundancy and inconsistency
 - c. Class imbalance
 - d. Underfitting
- iv) Which discretization method uses class labels to create intervals?
- a. Equal-width discretization
 - b. Entropy-based discretization
 - c. Equal-frequency discretization
 - d. Random discretization
- v) Which compression technique reduces data size without losing information?
- a. Autoencoders
 - b. PCA
 - c. Run-Length Encoding
 - d. Linear Discriminant Analysis
- vi) Which of the following best distinguishes supervised learning from unsupervised learning?
- a. Data size
 - b. Model complexity
 - c. Presence of labeled data
 - d. Feature dimensionality
- vii) Which task involves predicting a continuous output variable?
- a. Classification
 - b. Clustering
 - c. Regression
 - d. Association rule mining
- viii) Why is stratified sampling used while splitting datasets?
- a. To reduce training time
 - b. To maintain class distribution in splits
 - c. To increase model accuracy
 - d. To remove noisy data

- ix) What problem does label encoding primarily address?
- Handling missing values
 - Normalizing numerical data
 - Converting categorical labels into numeric form
 - Removing outliers

- x) What is the main purpose of cross-validation?
- To increase dataset size
 - To tune hyperparameters
 - To assess model generalization
 - To remove bias

2. a) Define data normalization. Why is it important in data preprocessing? (3)

b) Perform data smoothing using binning techniques on the following dataset. Use 3 bins and apply both equal-width binning and equal-frequency binning separately for smoothing by:

- bin means
- bin medians
- bin boundaries

Data: ~~12, 17, 25, 30, 18, 14, 9, 21, 16, 27, 5, 10, 24, 19~~ (4)

c) Apply the following normalization techniques to scale the above data:

- Min-Max normalization to the range [0, 1]
 - Z-Score normalization
 - Decimal Scaling
- (3)

3. a) Compute the Pearson and Spearman correlation coefficients between features, F_1 and F_2 for the given set of objects, and interpret the nature of the correlation between them. (4)

Object →	O_1	O_2	O_3	O_4	O_5
F_1	15	25	33	27	47
F_2	14	35	42	33	56

b) Explain Segmentation by Natural Partitioning (3-4-5 Rule) and apply it to iteratively partition the attribute values in the range -250 to 6,800 into sub-intervals, forming a hierarchical tree with up to three labels, where the root is considered as Label-1. Treat the data below the 5th percentile and above the 95th percentile separately, assuming the 5th percentile value is -125 and the 95th percentile value is 4,800. (6)